QF620 Stochastic Modelling in Finance — Session 4 Reference Cheatsheet

Stochastic Integrals and Itô's Formula: stochastic models and SDEs, Riemann vs stochastic integrals, properties and Itô isometry, box calculus, Itô's formula with and without time

Built from Session 4 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Session 4: Stochastic Integrals and Itô's Formula" (Tee Chyng Wen, SMU, 23 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. From this session you need: reading an SDE in differential and integral form; the left-endpoint definition of the stochastic integral; its mean, variance (Itô isometry) and normality for deterministic integrands; the box-calculus multiplication table; and applying Itô's formula, with and without explicit time dependence, to functions such as X_t^2, e^{X_t+t} and $\ln S_t$.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
W_t	Brownian motion	random	Session 3
S_t	stock price process	random	[S2]
μ, $\mu(t, S_t)$, μ_t	drift coefficient	number / function	expected change per unit time [S2–S3]
σ, $\sigma(t, S_t)$, σ_t	volatility (diffusion) coefficient	number / function	size of the random part [S2–S3]
$\mathrm{d}t$, $\mathrm{d}W_t$, $\mathrm{d}S_t$	infinitesimal changes	differentials	[S3]
Δt, ΔW_t	small finite changes	increments	$\Delta W_t = W_{t+\Delta t} - W_t$
P_n	partition $0 = t_0 < t_1 < \cdots < t_n = T$	set	[S6]
I_T	stochastic integral $\int_0^T f(u, W_u)\,\mathrm{d}W_u$	random	[S9]
f, g	integrand / function of the process	function	
f', f''	first, second derivative in the state variable	function	[S20]
g_t, g_x, g_{xx}	partial derivatives of $g(t, x)$	function	[S23]
X_t, Y_t	Itô processes	random	[S22]
$V[\cdot]$	variance	number	

Notation trap: in $g(t, X_t)$ the subscript in g_t means $\partial g/\partial t$ while the subscript in X_t means "at time t". f' always differentiates with respect to the state (W_t, S_t or X_t), never time.

0. Topic map

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PREREQUISITES
S3  Brownian motion: independent N(0, t−s) increments, nowhere differentiable, martingales
S1  Taylor series; S2 conditional-expectation rules (independence, taking out what is known)
  |
  ▼
SESSION 4: STOCHASTIC INTEGRALS AND ITÔ'S FORMULA
├─ MODELS  [S2–S3]
|   ├─ S_t = S₀ + μt + σW_t ~ N(S₀+μt, σ²t)                                  [S2]
|   └─ Integral form ⇔ differential form dS = μ dt + σ dW; dS = μ(t,S)dt + σ(t,S)dW  [S3]
|
├─ STOCHASTIC INTEGRAL  [S4–S12]
|   ├─ S_{t+Δt} = S_t + μΔt + σ(W_{t+Δt} − W_t): deterministic + stochastic   [S4]
|   ├─ dS/dt impossible: W nowhere differentiable                             [S5]
|   ├─ Riemann integral: Σ f(x_i)(t_i − t_{i−1}), any x_i in the interval      [S6–S7]
|   ├─ Stochastic integral: Σ f(t_{i−1})(W_{t_i} − W_{t_{i−1}}): left endpoint (previsible)  [S8]
|   ├─ Properties: mean 0, Itô isometry, normal if f deterministic            [S9]
|   ├─ Proofs of mean 0 and isometry                                          [S10–S11]
|   └─ Example: ∫₀ᵀ W dW: mean 0, variance T²/2                               [S12]
|
├─ BOX CALCULUS  [S13–S14]
|   ├─ Strong law of large numbers ⇒ (ΔW)² ≈ Δt                               [S13]
|   └─ dt·dt = 0, dt·dW = 0, dW·dW = dt; (dX)² = σ²dt                         [S14]
|
└─ ITÔ'S FORMULA  [S15–S23]
    ├─ History: Brown, Einstein, Wiener 1923, Itô 1942; why we need it      [S15–S18]
    ├─ Ordinary Taylor: dx = f′ dt;  stochastic: dX = f′ dW + ½f″ dt         [S19–S20]
    ├─ For S with dS = μ dt + σ dW: dX = (μf′ + ½σ²f″)dt + σf′ dW            [S21]
    ├─ Itô's formula Y = f(X_t); example Y = X²                             [S22]
    └─ Itô's formula with time Y = g(t, X_t); example Y = e^{X+t}           [S23]
  |
  ▼
LATER: GBM dS = μS dt + σS dW solved via d ln S; Black–Scholes PDE by Itô + hedging; Girsanov
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Dependency logic:

- Brownian paths have no derivative, so an SDE $\mathrm{d}S = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$ is shorthand for an *integral* equation; the stochastic integral is defined with left endpoints so the integrand is known before the increment happens.
- Independent, mean-zero increments give the two key properties: $\mathbb{E}[\int f\,\mathrm{d}W] = 0$ and the Itô isometry.
- $(\Delta W)^2$ behaves like Δt, not like $(\Delta t)^2$, so second-order Taylor terms survive. That extra $\frac{1}{2}f''\,\mathrm{d}t$ term is Itô's formula, and it is the source of every $-\frac{1}{2}\sigma^2$ seen in Sessions 2–3.

1 Concept entries

1.1 Building stochastic models [S2]

W_t alone is a poor stock model: we must control its **drift** and its **volatility**:

$$S_t = S_0 + \mu t + \sigma W_t, \qquad \mu \in \mathbb{R},\ \sigma \in \mathbb{R}.$$

where

S_t	stock price at time t.
S_0	stock price today.
μt	drift: deterministic trend, μ per unit time.
σW_t	random part: Brownian motion scaled by volatility σ.

$$S_t \sim N\big(S_0 + \mu t,\ \sigma^2 t\big), \qquad \mathbb{E}[S_t] = S_0 + \mu t, \qquad V[S_t] = V[\sigma W_t] = \sigma^2 t.$$

where

$\mathbb{E}[S_t]$	$\mathbb{E}[W_t] = 0$, so only the deterministic part survives.
$V[S_t]$	constants do not add variance; $V[\sigma W_t] = \sigma^2 V[W_t] = \sigma^2 t$.

[Beyond slides] Weakness: normally distributed, so S_t can go negative (arithmetic / Bachelier model).

1.2 Integral and differential forms; SDEs [S3]

$$S_t = S_0 + \int_0^t \sigma\,\mathrm{d}W_u + \int_0^t \mu\,\mathrm{d}u \qquad \Longleftrightarrow \qquad \mathrm{d}S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

where

$\int_0^t \mu\,\mathrm{d}u$	accumulated drift (an ordinary integral) $= \mu t$.
$\int_0^t \sigma\,\mathrm{d}W_u$	accumulated random shocks (a stochastic integral) $= \sigma W_t$.
u	dummy integration variable (time).
$\mathrm{d}S_t$	infinitesimal change of S over $[t, t + \mathrm{d}t]$.
$\mu\,\mathrm{d}t$	drift part of the change.
$\sigma\,\mathrm{d}W_t$	diffusion part: $\mathrm{d}W_t$ is the Brownian increment. The differential form is *shorthand* for the integral form.

In most models the coefficients depend on the current state and time only:

$$\mathrm{d}S_t = \mu(t, S_t)\,\mathrm{d}t + \sigma(t, S_t)\,\mathrm{d}W_t.$$

where

$\mu(t, S_t)$	drift coefficient: a function of time and the current price.
$\sigma(t, S_t)$	volatility coefficient. *[Beyond slides]* GBM: $\mu(t, S) = \mu S$, $\sigma(t, S) = \sigma S$.

With constant μ, σ the SDE is solved by $S_t = S_0 + \mu t + \sigma W_t$.

△**!** [S3] says "a stochastic process is a continuous process that can be written" in this form. That describes an *Itô process* (diffusion); many stochastic processes (e.g. with jumps) are not of this form.

1.3 Deterministic and stochastic components [S4–S5]

$$\underbrace{S_{t+\Delta t}}_{\text{future price}} = \underbrace{S_t}_{\text{present price}} + \underbrace{\mu(t, S_t)\Delta t}_{\text{deterministic}} + \underbrace{\sigma(t, S_t)\big(W_{t+\Delta t} - W_t\big)}_{\text{stochastic}}$$

where

Δt	small time step.
$\mu(t, S_t)\Delta t$	drift over the step.
$W_{t+\Delta t} - W_t$	Brownian increment $\sim N(0, \Delta t)$, independent of everything up to t.
$\sigma(t, S_t)$	scales the random increment.

Slide question [S4]: how is the drift "deterministic" when S_t is random? *[Beyond slides]*: $\mu(t, S_t)$ is evaluated at the *start* of the step, where S_t is already known. Given information at t, the drift over $[t, t + \Delta t]$ is a known number; only the Brownian increment is unknown.

Letting $\Delta t \to 0^+$ gives the **SDE** $\mathrm{d}S_t = \mu(t, S_t)\,\mathrm{d}t + \sigma(t, S_t)\,\mathrm{d}W_t$ [S5]. We cannot write

$$\frac{\mathrm{d}S_t}{\mathrm{d}t} = \mu(t, S_t) + \sigma(t, S_t)\frac{\mathrm{d}W_t}{\mathrm{d}t},$$

where

$\frac{\mathrm{d}W_t}{\mathrm{d}t}$	does not exist: Brownian motion is nowhere differentiable (Session 3), so the SDE must be read as an integral equation.

1.4 Integral form and the Riemann integral [S6–S7]

$$S_T = S_0 + \underbrace{\int_0^T \mu(u, S_u)\,\mathrm{d}u}_{\text{Riemann integral}} + \underbrace{\int_0^T \sigma(u, S_u)\,\mathrm{d}W_u}_{\text{stochastic integral}}.$$

where

T	horizon (maturity).
$\int \mu\,\mathrm{d}u$	integral against time: classical.
$\int \sigma\,\mathrm{d}W_u$	integral against Brownian increments: needs a new definition.

Riemann integral [S6–S7]: for a partition $P_n = \{t_0 = 0, t_1, \ldots, t_n = T\}$, f is Riemann integrable if

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i) \times (t_i - t_{i-1}), \qquad x_i \in [t_{i-1}, t_i]$$

converges (to the same limit for every choice of x_i).

where

$t_i - t_{i-1}$	width of the i-th sub-interval.
x_i	any point inside the i-th sub-interval (left, right, middle: all give the same limit for a regular f).
$f(x_i)(t_i - t_{i-1})$	area of the i-th rectangle.
$\lim_{n\to\infty}$	refine the partition: the sum of rectangles tends to the area under the curve.

1.5 The stochastic integral and previsibility [S8]

$$\int_0^T f(t)\,\mathrm{d}W_t = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_{i-1}) \times \left(W_{t_i} - W_{t_{i-1}}\right).$$

where

$f(t_{i-1})$	integrand evaluated at the **left endpoint** of each interval.
$W_{t_i} - W_{t_{i-1}}$	Brownian increment over the interval, playing the role of the width $t_i - t_{i-1}$.
lim	exists "in an appropriate sense" (mean square, L^2 *[Beyond slides]*), not path by path.

- The result is a **random variable**, not a number as with a Riemann integral.
- **Previsibility:** $f(x_i)$ must be known at the *beginning* of each interval, so $x_i = t_{i-1}$. Financially: you choose your position before the price moves. Other choices give different answers (e.g. midpoint gives the Stratonovich integral *[Beyond slides]*), because W is too rough for the choice to wash out.

1.6 Properties of stochastic integrals [S9]

For $I_T = \int_0^T f(u, W_u)\,\mathrm{d}W_u$:

$$(1)\quad \mathbb{E}[I_T] = 0$$

$$(2)\quad \mathbb{E}[I_T^2] = \mathbb{E}\Big[\Big(\int_0^T f(u,W_u)\,\mathrm{d}W_u\Big)^2\Big] = \mathbb{E}\Big[\int_0^T f(u,W_u)^2\,\mathrm{d}u\Big]$$

$$(3)\quad f \text{ deterministic } \Rightarrow\ I_T \sim N\Big(0,\ \int_0^T f(u)^2\,\mathrm{d}u\Big)$$

$$(4)\quad \mathbb{E}\Big[\int_0^T f(u)\,\mathrm{d}W_u \times \int_0^T g(s)\,\mathrm{d}W_s\Big] = \mathbb{E}\Big[\int_0^T f(u)g(u)\,\mathrm{d}u\Big]$$

$$(5)\quad \text{Itô isometry:}\quad \mathbb{E}\Big[\Big(\int_0^T X_t\,\mathrm{d}W_t\Big)^2\Big] = \mathbb{E}\Big[\int_0^T X_t^2\,\mathrm{d}t\Big]$$

where

$f(u,W_u),\ X_t$	integrand: may depend on time and the path so far (adapted / previsible).
$\mathbb{E}[I_T] = 0$	stochastic integrals have zero mean: each term is (known) × (mean-zero increment).
$\mathbb{E}[I_T^2]$	since the mean is 0, this is the **variance** of I_T.
$\int f^2\,\mathrm{d}u$	the "dW" becomes "du" after squaring and taking expectation, because $\mathbb{E}[(\Delta W)^2] = \Delta t$.
(3)	a deterministic integrand gives a sum of independent normals, hence normal.
(4)	covariance of two stochastic integrals against the same W; set $g = f$ to recover (2).
(5)	same statement as (2), in general notation.

△! [S9] Property (4) as written (with $f(u)$, $g(s)$) needs deterministic or adapted integrands; property (3) needs a deterministic f (with a random integrand I_T is usually not normal, e.g. $\int W\,\mathrm{d}W$).

1.7 Proofs of the mean and the isometry [S10–S11]

Proof of (1) [S10]:

$$\mathbb{E}\Big[\int_0^T f(t,W_t)\,\mathrm{d}W_t\Big] = \lim_{n\to\infty}\sum_{i=1}^n \mathbb{E}\big[f(t_{i-1},W_{t_{i-1}})\big] \times \mathbb{E}\big[W_{t_i} - W_{t_{i-1}}\big] = \lim_{n\to\infty}\sum_{i=1}^n \mathbb{E}\big[f(t_{i-1},W_{t_{i-1}})\big] \times 0 = 0.$$

where

$f(t_{i-1}, W_{t_{i-1}})$	depends only on the path up to t_{i-1}.
$W_{t_i} - W_{t_{i-1}}$	independent of $W_{t_{i-1}}$ (independent increments), so the expectation of the product factorises.
$\mathbb{E}[W_{t_i} - W_{t_{i-1}}] = 0$	Brownian increments have mean zero.

Proof of (2) [S11]: write $\Delta W_{t_{i-1}} = W_{t_i} - W_{t_{i-1}}$. Cross terms vanish:

$$\mathbb{E}\big[f(t_{i-1},W_{t_{i-1}})f(t_{j-1},W_{t_{j-1}})\,\Delta W_{t_{i-1}}\Delta W_{t_{j-1}}\big] = \begin{cases} \mathbb{E}\big[f(t_{i-1},W_{t_{i-1}})^2\big] \times (t_i - t_{i-1}) & i = j \\ 0 & i \neq j \end{cases}$$

where

$i \neq j$	say $i > j$: the later increment $\Delta W_{t_{i-1}}$ is independent of everything else in the product and has mean 0.
$i = j$	$\mathbb{E}[(\Delta W)^2] = t_i - t_{i-1}$ and the increment is independent of $f(t_{i-1},\cdot)$.

Hence

$$\mathbb{E}\Big[\Big(\int_0^T f\,\mathrm{d}W\Big)^2\Big] = \lim_{n\to\infty}\sum_{i=1}^n \mathbb{E}\big[f(t_{i-1},W_{t_{i-1}})^2\big](t_i - t_{i-1}) = \mathbb{E}\Big[\int_0^T f(t,W_t)^2\,\mathrm{d}t\Big].$$

where

$\sum_{i,j} \to \sum_i$	only the diagonal terms survive.
$\sum_i (\cdot)(t_i - t_{i-1})$	a Riemann sum, converging to $\int f^2\,\mathrm{d}t$.

△! [S10–S11] swap $\mathbb{E}$ and lim without comment; this is justified by mean-square convergence of the sums *[Beyond slides]*.

1.8 Example: $\int_0^T W_t \, dW_t$ [S12]

Worked here (*[Beyond slides]*):

$$\mathbb{E}[I_T] = 0, \qquad V[I_T] = \mathbb{E}[I_T^2] = \mathbb{E}\Big[\int_0^T W_t^2 \, dt\Big] = \int_0^T \mathbb{E}[W_t^2] \, dt = \int_0^T t \, dt = \frac{T^2}{2}.$$

where

$I_T = \int_0^T W_t \, dW_t$	integrand $f(t, W_t) = W_t$ (random, adapted).
$\mathbb{E}[I_T] = 0$	property (1).
$\mathbb{E}[I_T^2]$	Itô isometry, property (2).
$\mathbb{E}[W_t^2] = t$	variance of Brownian motion (Fubini lets $\mathbb{E}$ pass inside $\int dt$).

[Beyond slides] Closed form via Itô's formula on W_t^2 (§1.12): $\int_0^T W_t \, dW_t = \frac{1}{2}W_T^2 - \frac{1}{2}T$. Check: mean $\frac{1}{2}T - \frac{1}{2}T = 0$; variance $\frac{1}{4}V[W_T^2] = \frac{1}{4} \cdot 2T^2 = \frac{T^2}{2}$ ✓. Ordinary calculus would wrongly give $\frac{1}{2}W_T^2$.

1.9 Box calculus [S13–S14]

Strong law of large numbers [S13]:

$$\mathbb{P}\Big(\lim_{n\to\infty} \frac{X_1 + X_2 + \cdots + X_n}{n} = \mu\Big) = 1; \qquad X_i = \pm 1 \text{ equally likely } \Rightarrow \lim_{n\to\infty} \frac{X_1 + \cdots + X_n}{n} = 0.$$

where

X_i	i.i.d. random variables with common mean μ.
$\frac{1}{n}\sum X_i$	sample average.
$\mathbb{P}(\dots) = 1$	the average converges to the true mean "almost surely" (with probability 1).

△**!** [S13] writes "$n \to \infty \Rightarrow \mathbb{P}(\text{average} = \mu) = 1$"; the limit belongs inside the probability.

Why $(\Delta W)^2 \approx \Delta t$ *[Beyond slides]*: split $[0, T]$ into n steps. $\sum_i (\Delta W_i)^2 = \frac{T}{n}\sum_i Z_i^2$ with Z_i i.i.d. $N(0,1)$; by the law of large numbers $\frac{1}{n}\sum Z_i^2 \to \mathbb{E}[Z^2] = 1$, so the sum $\to T$ (the *quadratic variation*). Each $(\Delta W)^2$ contributes Δt on average, with relative fluctuations that vanish in the limit. So $(\Delta W_t)^2$ is of order Δt and ΔW_t of order $\sqrt{\Delta t}$.

Rules [S14]: under $\Delta t \to 0$,

$$\Delta t \to dt, \quad (\Delta t)^2 \to (dt)^2 = 0, \quad \Delta W_t \to dW_t, \quad (\Delta W_t)^2 \to dW_t^2 = dt, \quad (\Delta t \cdot \Delta W_t) \to dt \cdot dW_t = 0.$$

where

$(dt)^2 = 0$	order Δt^2: negligible compared with dt.
$dW_t^2 = dt$	order Δt: *not* negligible; this is the key rule.
$dt \, dW_t = 0$	order $\Delta t^{3/2}$: negligible.

$\cdot$	dt	dW_t
dt	0	0
dW_t	0	dt

Example [S14]:

$$dX_t = \mu \, dt + \sigma \, dW_t \quad \Rightarrow \quad (dX_t)^2 = \mu^2 (dt)^2 + 2\mu\sigma \, dt \, dW_t + \sigma^2 (dW_t)^2 = \sigma^2 \, dt.$$

where

$(dX_t)^2$	square of the increment, expanded like $(a+b)^2$.
$\mu^2(dt)^2$, $2\mu\sigma \, dt \, dW_t$	both zero by the table.
$\sigma^2 \, dt$	the only survivor: only the volatility matters for $(dX)^2$.

[Beyond slides] Two Brownian motions with correlation ρ: $dW^1 \, dW^2 = \rho \, dt$.

1.10 History and motivation [S15–S18]

- Brown (1827) pollen; Einstein, Perrin (early 1900s); Wiener (1923) built probability measures on path space with Lebesgue integrals; **Kiyosi Itô** (1942) built the stochastic integral and stochastic differential equations from scratch [S15–S16].

- **Why Itô's formula** [S17]: theoretically, to manipulate stochastic integrals with chain/product rules instead of going back to the definition; practically, an option price is a function of the stock price process, so we need the dynamics of a function of a stochastic process.
- Quotations [S18]: Hamming (computing for insight), Föllmer (Itô calculus transformed quantitative finance).

1.11 Deriving Itô's formula [S19–S21]

Ordinary calculus [S19]: Taylor expansion with $x = f(t)$:

$$\Delta x = f(t+\Delta t) - f(t) = f'(t)\Delta t + \frac{1}{2!}f''(t)(\Delta t)^2 + \frac{1}{3!}f'''(t)(\Delta t)^3 + \cdots \xrightarrow{\Delta t \to 0^+} \frac{\mathrm{d}x}{\mathrm{d}t} = f'(t).$$

where

f', f'', f'''	derivatives of f with respect to t.
$(\Delta t)^k,\ k \geq 2$	vanish faster than Δt, so only the first-order term remains.

Function of Brownian motion [S20]: with $X_t = f(W_t)$,

$$\Delta X_t = f'(W_t)\Delta W_t + \frac{1}{2!}f''(W_t)(\Delta W_t)^2 + \cdots \xrightarrow{\Delta t \to 0^+} \mathrm{d}X_t = f'(W_t)\,\mathrm{d}W_t + \underbrace{\tfrac{1}{2}f''(W_t)\,\mathrm{d}t}_{\text{extra term}}.$$

where

$f'(W_t),\ f''(W_t)$	derivatives of f with respect to its argument, evaluated at W_t.
$(\Delta W_t)^2 \to \mathrm{d}t$	box calculus: the second-order term is of order $\mathrm{d}t$ and survives.
$\frac{1}{2}f''\,\mathrm{d}t$	the Itô correction, absent in ordinary calculus. Third and higher orders, $(\mathrm{d}W)^3 \sim \mathrm{d}t^{3/2}$, vanish.

Function of a stock [S21]: $\Delta S_t = \mu\Delta t + \sigma\Delta W_t$ and $X_t = f(S_t)$:

$$\Delta X_t = f'(S_t)\Delta S_t + \tfrac{1}{2}f''(S_t)(\Delta S_t)^2 + \cdots, \qquad (\Delta S_t)^2 = \mu^2(\Delta t)^2 + 2\mu\sigma\,\Delta t\,\Delta W_t + \sigma^2(\Delta W_t)^2 \to \sigma^2\,\mathrm{d}t,$$

where

ΔS_t	stock increment over the step.
$(\Delta S_t)^2$	only the $\sigma^2(\Delta W)^2$ term survives, giving $(\mathrm{d}S_t)^2 = \sigma^2\mathrm{d}t$.

so

$$\mathrm{d}X_t = f'(S_t)\,\mathrm{d}S_t + \tfrac{1}{2}f''(S_t)\sigma^2\,\mathrm{d}t = \Big(\mu f'(S_t) + \frac{\sigma^2}{2}f''(S_t)\Big)\mathrm{d}t + \sigma f'(S_t)\,\mathrm{d}W_t.$$

where

$f'(S_t)\,\mathrm{d}S_t$	first-order (ordinary chain rule) term.
$\frac{1}{2}f''(S_t)\sigma^2\,\mathrm{d}t$	second-order Itô term.
$\mu f' + \frac{1}{2}\sigma^2 f''$	drift of X: chain-rule drift plus convexity correction.
$\sigma f'$	volatility of X: the stock's volatility times the sensitivity f' (a "delta").

1.12 Itô's formula (function of a stochastic process) [S22]

If $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t$ and $f : \mathbb{R} \to \mathbb{R}$ with f, f', f'' continuous, then $Y_t = f(X_t)$ satisfies

$$\mathrm{d}Y_t = \Big(\mu_t f'(X_t) + \frac{1}{2}\sigma_t^2 f''(X_t)\Big)\mathrm{d}t + \sigma_t f'(X_t)\,\mathrm{d}W_t.$$

where

X_t	an Itô process (the underlying).
$\mu_t,\ \sigma_t$	drift and volatility of X at time t (may be random, but known at t).
f	smooth function (C^2); f', f'' = first and second derivatives.
$Y_t = f(X_t)$	new process, e.g. an option value as a function of the stock.
$\frac{1}{2}\sigma_t^2 f''(X_t)$	Itô correction from $(\mathrm{d}X)^2 = \sigma_t^2\mathrm{d}t$.

Slide example [S22] (worked here, *[Beyond slides]*): $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $Y_t = X_t^2$. Then $f(x) = x^2$, $f' = 2x$, $f'' = 2$:

$$\mathrm{d}Y_t = \big(2\mu X_t + \sigma^2\big)\mathrm{d}t + 2\sigma X_t\,\mathrm{d}W_t.$$

where

$2\mu X_t$	ordinary chain-rule drift $\mu f'$.
σ^2	Itô term $\frac{1}{2}\sigma^2 \cdot 2$.
$2\sigma X_t$	volatility $\sigma f'$.

[Beyond slides] With $\mu = 0, \sigma = 1$ ($X = W$): $\mathrm{d}(W_t^2) = \mathrm{d}t + 2W_t\,\mathrm{d}W_t$, so $\int_0^T W\,\mathrm{d}W = \frac{1}{2}W_T^2 - \frac{1}{2}T$ and $W_t^2 - t = 2\int_0^t W\,\mathrm{d}W$ is a martingale (Session 3).

1.13 Itô's formula with time [S23]

If $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t$ and $g : \mathbb{R}^2 \to \mathbb{R}$ with g, g_t, g_x, g_{xx} continuous, then $Y_t = g(t, X_t)$ satisfies

$$\mathrm{d}Y_t = \Big[g_t(t,X_t) + \mu_t g_x(t,X_t) + \frac{1}{2}\sigma_t^2 g_{xx}(t,X_t)\Big]\mathrm{d}t + \sigma_t g_x(t,X_t)\,\mathrm{d}W_t.$$

where

$g(t,x)$	function of time and the state (e.g. an option price $V(t,S)$).
$g_t = \dfrac{\partial g}{\partial t}$	sensitivity to the passage of time ("theta").
$g_x = \dfrac{\partial g}{\partial x}$	sensitivity to the state ("delta").
$g_{xx} = \dfrac{\partial^2 g}{\partial x^2}$	curvature in the state ("gamma"); produces the Itô term.
no g_{tt}, g_{tx}	they multiply $(\mathrm{d}t)^2$ and $\mathrm{d}t\,\mathrm{d}W$, which are zero.

Slide example [S23] (worked here, *[Beyond slides]*): $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $Y_t = e^{X_t + t}$. Then $g = g_t = g_x = g_{xx} = e^{x+t} = Y$:

$$\mathrm{d}Y_t = \Big(1 + \mu + \frac{1}{2}\sigma^2\Big)Y_t\,\mathrm{d}t + \sigma Y_t\,\mathrm{d}W_t.$$

where

$1 \cdot Y_t$	from g_t (the explicit $+t$ in the exponent).
μY_t	from μg_x.
$\frac{1}{2}\sigma^2 Y_t$	Itô term from g_{xx}.
σY_t	volatility: proportional to Y, so Y is a geometric Brownian motion.

GBM via Itô *[Beyond slides]*: $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$, $f = \ln$ ($f' = 1/S$, $f'' = -1/S^2$):

$$\mathrm{d}(\ln S_t) = \Big(\mu - \frac{1}{2}\sigma^2\Big)\mathrm{d}t + \sigma\,\mathrm{d}W_t \quad \Rightarrow \quad S_t = S_0 \exp\Big[\Big(\mu - \frac{1}{2}\sigma^2\Big)t + \sigma W_t\Big].$$

where

μS_t, σS_t	drift and volatility proportional to the price (percentage returns).
$\mu - \dfrac{1}{2}\sigma^2$	log-drift: $\mu \cdot \frac{1}{S} \cdot S + \frac{1}{2}\sigma^2 S^2 \cdot (-\frac{1}{S^2})$. This is the $r - \frac{1}{2}\sigma^2$ of Session 2 when $\mu = r$.
$\sigma\,\mathrm{d}W_t$	$\sigma S \cdot \frac{1}{S}$: constant volatility in log space.

1.14 Implementation *[Beyond slides]* (tested)

```
import numpy as np
rng = np.random.default_rng(1)
paths, steps, T = 200_000, 500, 2.0
dt = T / steps
dW = np.sqrt(dt) * rng.standard_normal((paths, steps))
W = np.cumsum(dW, axis=1)
W_left = np.hstack([np.zeros((paths, 1)), W[:, :-1]])   # left end point (Ito)
I = np.sum(W_left * dW, axis=1)                         # sum W_{t_{i-1}} dW_i
print(round(I.mean(), 3), round(I.var(), 3), T**2/2)    # -0.0 2.004 2.0
print(round(np.mean(np.abs(I - (0.5*W[:, -1]**2 - 0.5*T))), 3))  # 0.05, shrinks as steps grow
QV = np.sum(dW**2, axis=1)
print(round(QV.mean(), 3), round(QV.std(), 3))          # 2.0 0.127: sum of dW^2 ~ T
```

2 Comparison tables

2.1 Riemann vs stochastic integral

	Riemann $\int f(t)\,\mathrm{d}t$ [S6–S7]	Stochastic $\int f\,\mathrm{d}W_t$ [S8]
Sum	$\sum f(x_i)(t_i - t_{i-1})$	$\sum f(t_{i-1})(W_{t_i} - W_{t_{i-1}})$
Evaluation point	any $x_i \in [t_{i-1}, t_i]$	left endpoint only (previsible)
Result	a number	a random variable
Limit	ordinary	mean square *[Beyond slides]*
Mean	—	0
Second moment	—	$\mathbb{E}\int f^2\,\mathrm{d}t$ (Itô isometry)

2.2 Ordinary vs Itô calculus

	Ordinary [S19]	Itô [S20–S23]
Second-order term	$(\Delta t)^2 \to 0$	$(\Delta W)^2 \to \mathrm{d}t$ survives
Chain rule	$\mathrm{d}f(x) = f'\,\mathrm{d}x$	$\mathrm{d}f(X) = f'\,\mathrm{d}X + \frac{1}{2}f''(\mathrm{d}X)^2$
$\mathrm{d}(W^2)$	$2W\,\mathrm{d}W$	$2W\,\mathrm{d}W + \mathrm{d}t$
$\int_0^T W\,\mathrm{d}W$	$\frac{1}{2}W_T^2$	$\frac{1}{2}W_T^2 - \frac{1}{2}T$
$\mathrm{d}\ln S$ for GBM	$\mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$	$(\mu - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma\,\mathrm{d}W$

2.3 Box calculus table [S14]

$\mathrm{d}t \cdot \mathrm{d}t$	0
$\mathrm{d}t \cdot \mathrm{d}W_t$	0
$\mathrm{d}W_t \cdot \mathrm{d}W_t$	$\mathrm{d}t$
$(\mu\,\mathrm{d}t + \sigma\,\mathrm{d}W)^2$	$\sigma^2\,\mathrm{d}t$
$\mathrm{d}W^1 \cdot \mathrm{d}W^2$	$\rho\,\mathrm{d}t$ (correlated BMs) *[Beyond slides]*

2.4 Itô's formula: the two versions

	$Y = f(X_t)$ [S22]	$Y = g(t, X_t)$ [S23]
Drift	$\mu_t f' + \frac{1}{2}\sigma_t^2 f''$	$g_t + \mu_t g_x + \frac{1}{2}\sigma_t^2 g_{xx}$
Volatility	$\sigma_t f'$	$\sigma_t g_x$
Needs	$f \in C^2$	g_t, g_x, g_{xx} continuous
Slide example	X^2: $(2\mu X + \sigma^2)\mathrm{d}t + 2\sigma X\,\mathrm{d}W$	e^{X+t}: $(1 + \mu + \frac{1}{2}\sigma^2)Y\mathrm{d}t + \sigma Y\mathrm{d}W$

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Models and SDEs [S2–S5]

- $S_t = S_0 + \mu t + \sigma W_t \sim N(S_0 + \mu t, \sigma^2 t)$.
  where μ = drift; σ = volatility; W_t = Brownian motion.
- $\mathrm{d}S_t = \mu(t, S_t)\,\mathrm{d}t + \sigma(t, S_t)\,\mathrm{d}W_t \iff S_T = S_0 + \int_0^T \mu\,\mathrm{d}u + \int_0^T \sigma\,\mathrm{d}W_u$.
  where first integral = Riemann; second = stochastic.

Stochastic integral [S8–S12]

- $\int_0^T f\,\mathrm{d}W = \lim \sum f(t_{i-1})(W_{t_i} - W_{t_{i-1}})$.

where $f(t_{i-1})$ = left endpoint (previsibility).

- $\mathbb{E}[\int f\,\mathrm{d}W] = 0$; $\mathbb{E}[(\int f\,\mathrm{d}W)^2] = \mathbb{E}[\int f^2\,\mathrm{d}t]$; f deterministic $\Rightarrow N(0, \int f^2 \mathrm{d}t)$.
  where Itô isometry; f = integrand.
- $\int_0^T W\,\mathrm{d}W = \frac{1}{2}W_T^2 - \frac{1}{2}T$: mean 0, variance $T^2/2$.
  where T = horizon.

Box calculus [S13–S14]

- $(\mathrm{d}t)^2 = 0$, $\mathrm{d}t\,\mathrm{d}W = 0$, $(\mathrm{d}W)^2 = \mathrm{d}t$; $(\mathrm{d}X)^2 = \sigma^2 \mathrm{d}t$ for $\mathrm{d}X = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$.
  where only $(\mathrm{d}W)^2$ survives.

Itô's formula [S20–S23]

- $\mathrm{d}f(W_t) = f'(W_t)\,\mathrm{d}W_t + \frac{1}{2}f''(W_t)\,\mathrm{d}t$.
  where f', f'' = derivatives in the state.
- $\mathrm{d}f(X_t) = \big(\mu_t f' + \frac{1}{2}\sigma_t^2 f''\big)\mathrm{d}t + \sigma_t f'\,\mathrm{d}W_t$.
  where $\mathrm{d}X = \mu_t \mathrm{d}t + \sigma_t \mathrm{d}W$.
- $\mathrm{d}g(t, X_t) = \big(g_t + \mu_t g_x + \frac{1}{2}\sigma_t^2 g_{xx}\big)\mathrm{d}t + \sigma_t g_x\,\mathrm{d}W_t$.
  where g_t, g_x, g_{xx} = partial derivatives in time, state, state twice.
- GBM: $\mathrm{d}S = \mu S \mathrm{d}t + \sigma S \mathrm{d}W \Rightarrow S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$ *[Beyond slides].*
  where apply Itô to $\ln S$.

4 Glossary

- **Box calculus:** multiplication table for $\mathrm{d}t$ and $\mathrm{d}W$ [S14].
- **Diffusion / volatility coefficient $\sigma(t, S_t)$:** scales $\mathrm{d}W$ [S3].
- **Drift coefficient $\mu(t, S_t)$:** scales $\mathrm{d}t$ [S3].
- **Geometric Brownian motion:** $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W$ *[Beyond slides].*
- **Itô process:** $\mathrm{d}X = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W$ [S22].
- **Itô isometry:** $\mathbb{E}[(\int X\,\mathrm{d}W)^2] = \mathbb{E}[\int X^2\,\mathrm{d}t]$ [S9].
- **Itô's formula / lemma:** chain rule for stochastic processes [S22–S23].
- **Partition:** $0 = t_0 < t_1 < \cdots < t_n = T$ [S6].
- **Previsibility:** integrand known at the start of each interval [S8].
- **Quadratic variation:** $\sum(\Delta W)^2 \to T$ *[Beyond slides].*
- **Riemann integral:** limit of rectangle sums [S6–S7].
- **SDE:** stochastic differential equation [S5].
- **Stochastic integral:** limit of left-endpoint sums against ΔW [S8].
- **Strong law of large numbers:** sample averages converge to the mean with probability 1 [S13].

5 Self-test

Questions

Q1 ([S2]). For $S_t = S_0 + \mu t + \sigma W_t$ give the distribution of S_t and $\mathbb{P}(S_t < 0)$.

Q2 ([S4–S5]). Why is the drift term "deterministic" over a step, and why can't we write $\mathrm{d}S/\mathrm{d}t$?

Q3 ([S8]). Why is the stochastic integral defined with left endpoints?

Q4 ([S9]). Find the distribution of $\int_0^T t\,\mathrm{d}W_t$.

Q5 ([S12]). Find the mean and variance of $\int_0^T W_t\,\mathrm{d}W_t$.

Q6 ([S14]). Compute $(\mathrm{d}X_t)^2$ for $\mathrm{d}X = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$ and $(\mathrm{d}S)^2$ for $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W$.

Q7 ([S22]). $\mathrm{d}X = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$, $Y = X^2$. Find $\mathrm{d}Y$.

Q8 ([S23]). Same X, $Y = e^{X+t}$. Find $\mathrm{d}Y$.

Q9. Use Itô's formula to show $\int_0^T W\,\mathrm{d}W = \frac{1}{2}W_T^2 - \frac{1}{2}T$.

Q10. Solve $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W$ and find $\mathbb{E}[S_t]$.

Solutions

S1. $N(S_0+\mu t, \sigma^2 t)$; $\mathbb{P}(S_t<0)=\Phi\big(-\frac{S_0+\mu t}{\sigma\sqrt{t}}\big)>0$: negative prices possible.

S2. $\mu(t,S_t)$ uses the price at the start of the step, already known at t. W is nowhere differentiable so $\mathrm{d}W/\mathrm{d}t$ does not exist; the SDE means the integral equation.

S3. Previsibility: the integrand (a trading position) must be fixed before the increment is revealed. Other evaluation points give different limits because W is rough.

S4. Deterministic integrand: $N(0, \int_0^T t^2\,\mathrm{d}t) = N(0, T^3/3)$.

S5. Mean 0; variance $\int_0^T \mathbb{E}[W_t^2]\,\mathrm{d}t = T^2/2$.

S6. $\sigma^2\,\mathrm{d}t$; $\sigma^2 S^2\,\mathrm{d}t$.

S7. $\mathrm{d}Y = (2\mu X+\sigma^2)\mathrm{d}t + 2\sigma X\,\mathrm{d}W$.

S8. $\mathrm{d}Y = (1+\mu+\frac{1}{2}\sigma^2)Y\,\mathrm{d}t + \sigma Y\,\mathrm{d}W$.

S9. $f(w)=w^2$: $\mathrm{d}(W^2) = 2W\,\mathrm{d}W + \mathrm{d}t$; integrate from 0 to T: $W_T^2 = 2\int W\,\mathrm{d}W + T$.

S10. $\mathrm{d}\ln S = (\mu-\frac{1}{2}\sigma^2)\mathrm{d}t + \sigma\,\mathrm{d}W$, so $S_t = S_0 e^{(\mu-\frac{1}{2}\sigma^2)t+\sigma W_t}$; $\mathbb{E}[S_t] = S_0 e^{(\mu-\frac{1}{2}\sigma^2)t} e^{\frac{1}{2}\sigma^2 t} = S_0 e^{\mu t}$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	$S_t = S_0 + \mu t + \sigma W_t$; mean and variance	1.1
S3	Integral and differential forms; state-dependent coefficients	1.2
S4	Deterministic and stochastic components; question on drift	1.3
S5	SDE; $\mathrm{d}S/\mathrm{d}t$ not feasible	1.3
S6	Integral form; Riemann integration definition	1.4
S7	Riemann integral as area	1.4
S8	Stochastic integral; previsibility	1.5
S9	Properties of stochastic integrals; Itô isometry	1.6
S10	Proof of (1)	1.7
S11	Proof of (2)	1.7
S12	Example $\int W\,\mathrm{d}W$	1.8
S13	Strong law of large numbers; $(\Delta W)^2 \sim \Delta t$	1.9
S14	Box calculus rules and table	1.9
S15	History	1.10
S16	Kiyosi Itô	1.10
S17	Why Itô's formula	1.10
S18	Quotations	1.10
S19	Ordinary Taylor expansion	1.11
S20	Taylor for $f(W_t)$; extra term	1.11
S21	Taylor for $f(S_t)$	1.11
S22	Itô's formula; example X^2	1.12
S23	Itô's formula with time; example e^{X+t}	1.13

Appendix B. Flag summary (△!)

1. [S3] "A stochastic process is a continuous process that can be written …": this defines an Itô process, not every stochastic process.
2. [S9] Property (3) needs a deterministic integrand; (4) as written needs deterministic or adapted f, g. (2) and (5) are the same statement.
3. [S10–S11] Exchange of $\mathbb{E}$ and lim is justified by mean-square convergence, not shown.
4. [S13] The strong law should read $\mathbb{P}(\lim_n \frac{1}{n}\sum X_i = \mu) = 1$. The real reason $(\Delta W)^2 \approx \Delta t$ is quadratic variation $\sum(\Delta W)^2 \to T$.
5. [S4, S12, S22, S23] Questions and examples are posed without worked answers; the solutions here are my own.